

Vector Spatial Analysis for Flood Response & Infrastructure Exposure in Sindh, Pakistan

Multi-Criteria Vector Overlay, Network Accessibility & Hydrological Vulnerability Analysis

Level: Advanced Undergraduate / Graduate

Software: QGIS 3.22+ / ArcGIS Pro

Estimated Time: 3 Hours

Target Projection: EPSG:32642 (UTM 42N)

1. Operational Scenario & Background

During severe monsoon events along the lower Indus River basin in **Sindh Province, Pakistan**, catastrophic flooding frequently disrupts road transport, isolates health centers, and leaves vulnerable rural populations cut off from disaster relief. The Provincial Disaster Management Authority (PDMA) Sindh requires a vector-based GIS assessment to identify critical infrastructure exposed to rising floodwaters.

In this laboratory exercise, you will act as a GIS Lead for emergency response. You will acquire, standardise, project, and query vector datasets representing administrative divisions, river networks, road infrastructure, population centers, and health facilities. Through spatial buffering, vector intersection, distance matrix generation, and attribute joins, you will deliver actionable spatial metrics for disaster response.

23 Districts

SINDH STUDY AREA

3 Flood Zones

1 KM, 3 KM, & 5 KM RISK BUFFERS

UTM Zone 42N

METRIC PROJECTING (EPSG:32642)

2. Primary Vector Datasets & Acquisition Matrix

All data must be gathered from open-access geospatial platforms. Verify geometry validity and projected coordinates upon import.

Dataset Name	Geometry	Key Attributes	Source Portal	Format & URL
Pakistan Admin Boundaries	Polygon	ADM1_EN (Province), ADM2_EN (District), ADM3_EN (Tehsil)	Humanitarian Data Exchange (HDX) / GADM	Shapefile (.shp) data.humdata.org
Indus River Mainstem & Tributaries	Line / Polygon	NAME, WATERWAY, HYDR_TYPE	Survey of Pakistan / OpenStreetMap / HDX	GeoJSON / Shapefile data.humdata.org
Sindh Road Network	Line	highway, ref, surface, lanes	OpenStreetMap via Overpass Turbo	GeoJSON overpass-turbo.eu
Healthcare Facilities	Point	amenity, name, facility_type, capacity	WHO / OCHA Pakistan Data Portal	CSV / Shapefile data.humdata.org
Settlements & Urban Centers	Point / Polygon	place, name, pop_est, district	Humanitarian OpenStreetMap Team (HOT)	GeoPackage export.hotosm.org

CRS Requirement Note: Data downloaded from global repositories typically uses geographic coordinates (EPSG:4326 - WGS 84). All distance and area measurements in this lab require metric units. You MUST reproject all datasets to **WGS 84 / UTM Zone 42N (EPSG:32642)** prior to performing buffer or intersection operations.

3. Step-by-Step Practical Instructions

Module 1: Vector Acquisition & Layer Standardization

- 1 Extract Administrative Boundaries:** Load the national Tehsil boundary vector file (pak_admbnda_adm3_30m.shp). Filter the dataset to isolate Sindh Province using the SQL query clause:

```
"ADM1_EN" = 'Sindh'
```

- 2 Execute Overpass Turbo Query for Infrastructure:** Open [Overpass Turbo](#), pan to Southern Pakistan (Dadu / Sukkur / Hyderabad corridor), and execute the following Overpass QL script:

```
[out:json][timeout:60];
(
  // Extract primary transportation infrastructure
  way["highway"~"trunk|primary|secondary"]({{bbox}});
  // Extract emergency medical facilities
  node["amenity"~"hospital|clinic"]({{bbox}});
);
out body;
>;
out skel qt;
```

- 3 Reprojection to Local UTM Metric Grid:** Export all extracted vector layers into a unified GeoPackage named Sindh_Flood_Analysis.gpkg . Set the target CRS explicitly to EPSG:32642 (WGS 84 / UTM Zone 42N) .

Module 2: Attribute Queries & Geometric Metric Computation

- 1 Tehsil Area Calculation:** Open the attribute table for the filtered Sindh Tehsil polygons. Enable editing, open the *Field Calculator*, and create a new field named Tehsil_SqKm (Type: Decimal / Real, Precision: 2). Apply the formula:

$$Area_{SqKm} = \$area / 1,000,000$$

- 2 Arterial Highway Isolation:** Create a subset layer of high-capacity arterial highways by executing an attribute expression on the road layer:

```
"highway" IN ('trunk', 'primary') AND "surface" NOT IN ('unpaved', 'dirt')
```

- 3 Health Center Classification:** Isolate functional District Headquarter (DHQ) hospitals and Basic Health Units (BHUs) into a dedicated point feature layer named sindh_health_facilities .

Module 3: Proximity Analysis & Spatial Overlays

- 1 Hydrological Multi-Ring Buffers:** Using the Indus River vector line dataset, generate multi-distance buffer polygons representing flood hazard bands:

- **High Hazard Zone:** 1,000 meters (1 km)
- **Moderate Hazard Zone:** 3,000 meters (3 km)
- **Extended Risk Zone:** 5,000 meters (5 km)

2 Inundated Road Network Intersect: Execute the *Intersection* tool (*Vector -> Geoprocessing Tools -> Intersection*) between the primary roads layer and the 3 km flood risk buffer polygon.

Open the resulting intersection attribute table and calculate the impacted segment lengths in kilometers:

$$Segment_Length_{Km} = \$length / 1,000$$

3 Point-in-Polygon Facility Aggregation: Run the *Count Points in Polygon* tool using the Tehsil polygon layer and the healthcare point feature class. Store the results in a new field named `Health_Count` .

Module 4: Accessibility & Distance Matrix Analysis

1 Identify Isolated Medical Infrastructure: Run a *Select by Location* query to find healthcare facility points that intersect with the 3 km flood hazard zone. Export the remaining unflooded hospitals as `active_health_centers` .

2 Distance Matrix Computation: Execute the *Distance Matrix* tool using settlement centroids as input origin points and `active_health_centers` as target destination points. Set the matrix type to **Linear Distance Matrix (1 to nearest 1 target)**.

4. Analytical Questions & Reporting Checklist

Answer the following questions using the layers generated during the analysis:

1. **Road Infrastructure Vulnerability:** What is the total length (\$ ext{in km}\$) of primary and trunk highways located within the 3 km flood buffer across Sindh Province? Which specific Tehsil incurs the highest road length exposure?

2. **Healthcare Inaccessibility:** How many health facilities fall directly into the High Hazard Zone (1 km buffer)? What proportion of the total healthcare infrastructure does this represent?

3. **Community Isolation Metrics:** Based on your Distance Matrix output, identify the top three settlement points that experience the greatest increase in straight-line distance to an operational (unflooded) health facility.

4. **CRS Impact Reflection:** Explain why performing this analysis in `EPSG:4326 (WGS 84)` without reprojecting to `EPSG:32642` would produce invalid length and buffer calculation results.

5. Expected Lab Deliverables & Rubric

Deliverable	Specifications & Requirements	Weight
1. Integrated GeoPackage Database	<code>Sindh_Flood_Analysis.gpkg</code> containing all reprojected vector layers (<code>EPSG:32642</code>), calculated metrics, clean attribute names, and zero null geometries.	30%
2. Cartographic Map Output (PDF)	High-resolution A4 map depicting Tehsil health density (choropleth), river hazard buffers, inundated vs operational roads, and cartographic elements (Legend, Scale bar in km, North Arrow, UTM Grid).	40%
3. Vector Metrics Summary Table	A structured CSV/Excel summary table detailing total road length inundation (\$ ext{in km}\$) and health facility counts aggregated by Tehsil and District.	30%

Submission Guidelines: Compress your final GeoPackage, PDF Map, and CSV Summary Table into a single zip file named `GIS402_Lab4_[YourID].zip` and upload to the course portal.